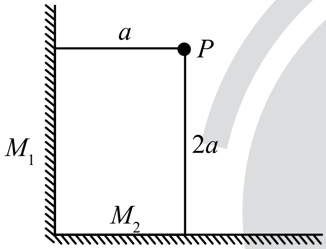
Ray Optics and Optical Instruments

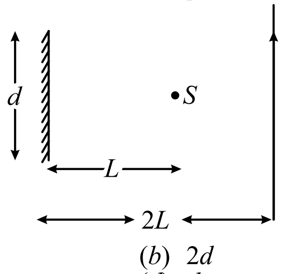
Single Correct Type Questions

1. Two plane mirrors M_1 and M_2 are at right angle to each shown. A point source 'P' is placed at 'a' and '$2a$' meter away from M_1 and M_2 respectively. The shortest distance between the images thus formed is : (Take $\sqrt{5} = 2.3$)
[31 Aug, 2021 (Shift-I)]



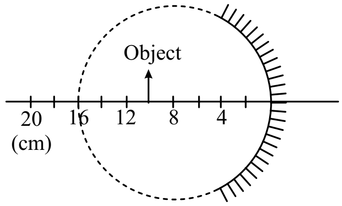
(*a*) $2\sqrt{10}\,a$ (*b*) $2.3\,a$
(*c*) $4.6\,a$ (*d*) $3\,a$

2. A point source of light, S is placed at a distance L in front of the centre of plane mirror of width d which is hanging vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror, at a distance $2\,L$ as shown below. The distance over which the man can see the image of the light source in the mirror:
[12 Jan, 2019 (Shift-I)]



(*a*) d (*b*) $2d$
(*c*) $3d$ (*d*) d

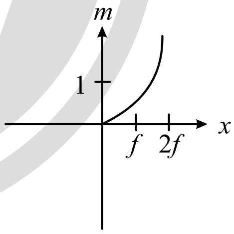
3.

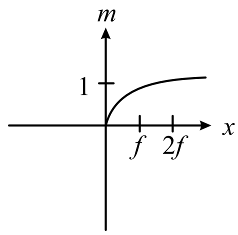


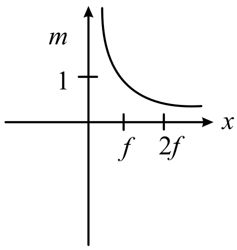
A spherical mirror is obtained as shown in the figure from a hollow glass sphere. if an object is positioned infront of the mirror, what will be the nature and magnification of the image of the object? (Figure drawn as schematic and not to scale) **[2 Sep, 2020 (Shift-I)]**

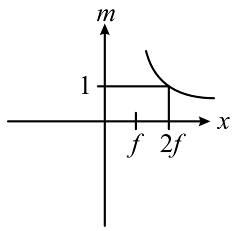
(*a*) Inverted, real and magnified
(*b*) Erect, virtual and unmagnified
(*c*) Inverted, real and unmagnified
(*d*) Erect, virtual and magnified

4. An object is gradually moving away from the focal point of a concave mirror along the axis of the mirror. The graphical representation of the magnitude of linear magnification (m) versus distance of the object from the mirror (x) is correctly given by (Graphs are drawn schematically and are not to scale)
[8 Jan, 2020 (Shift-II)]

(*a*)


(*b*)


(*c*)


(*d*)


5. A vessel of depth 'd' is half filled with oil of refractive index n_1 and the other half is filled with water of refractive index n_2. The apparent depth of this vessel when viewed from above will be **[13 Apr 2023 (Shift-I)]**

(*a*) $\dfrac{dn_1n_2}{(n_1+n_2)}$ (*b*) $\dfrac{d(n_1+n_2)}{2n_1n_2}$

(*c*) $\dfrac{dn_1n_2}{2(n_1+n_2)}$ (*d*) $\dfrac{2d(n_1+n_2)}{n_1n_2}$

6. Time taken by light to travel in two different materials A and B refractive indices μ_A and μ_B of same thickness is t_1 and t_2 respectively. If $t_2 - t_1 = 5 \times 10^{-10}$ s and the ratio of μ_A and μ_B is 1 : 2. Then, the thickness of material, in meter is: (Given υ_A and υ_B are velocities of light in A and B materials respectively.) **[25 July, 2022 (Shift-I)]**

(*a*) $5 \times 10^{-10}\,\upsilon_A$ m (*b*) 5×10^{-10} m
(*c*) 1.5×10^{-10} m (*d*) $5 \times 10^{-10}\,\upsilon_B$ m

7. Consider a light ray travelling in air is incident into a medium of refractive index $\sqrt{2n}$. The incident angle is twice that of refracting angle. Then, the angle of incidence will be **[27 June, 2022 (Shift-I)]**

(*a*) $\sin^{-1}\left(\sqrt{n}\right)$ (*b*) $\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$
(*c*) $\sin^{-1}\left(\sqrt{2n}\right)$ (*d*) $2\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

8. A ray of laser of a wavelength 630 nm is incident at an angle of 30º at the diamond- air interface. It is going from diamond to air. The refractive index diamond is 2.42 and that of air is 1. Choose the correct option.

[25 July, 2021 (Shift-I)]

(*a*) Angle of refraction is 53.4º
(*b*) Angle of refraction is 30º
(*c*) Angle of refraction is 24.41º
(*d*) Refraction is not possible

9. A vessel of depth 2 *h* is half filled with a liquid of refractive index $2\sqrt{2}$ and the upper half with another liquid of refractive index $\sqrt{2}$. The liquids are immiscible. The apparent depth of the inner surface of the bottom of vessel will be **[9 Jan, 2020 (Shift-I)]**

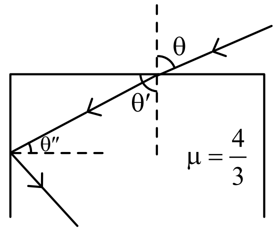
(*a*) $\frac{h}{\sqrt{2}}$ (*b*) $\frac{h}{2\left(\sqrt{2}+1\right)}$
(*c*) $\frac{3}{4}h\sqrt{2}$ (*d*) $\frac{h}{3\sqrt{2}}$

10. The speed of light in media 'A' and 'B' are 2.0×10^{10} cm/s and 1.5×10^{10} cm/s respectively. A ray of light enters from the medium *B* to *A* an incident angle 'θ'. If the ray suffers total internal reflection, then **[29 June, 2022 (Shift-II)]**

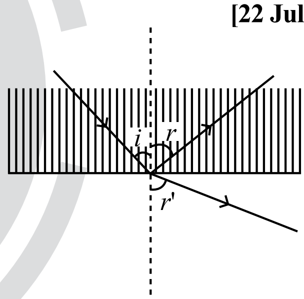
(*a*) $\theta = \sin^{-1}\left(\frac{3}{4}\right)$ (*b*) $\theta > \sin^{-1}\left(\frac{2}{3}\right)$
(*c*) $\theta < \sin^{-1}\left(\frac{3}{4}\right)$ (*d*) $\theta > \sin^{-1}\left(\frac{3}{4}\right)$

11. A ray of light entering from air into a denser medium of refractive index $\frac{4}{3}$, as shown in figure. The light ray suffers total internal reflection at the adjacent surface as shown. The maximum value of angle θ should be equal to : **[25 July, 2021 (Shift-II)]**

(*a*) $\sin^{-1}\frac{\sqrt{5}}{4}$ (*b*) $\sin^{-1}\frac{\sqrt{7}}{3}$
(*c*) $\sin^{-1}\frac{\sqrt{7}}{4}$ (*d*) $\sin^{-1}\frac{\sqrt{5}}{3}$



12. A ray of light passes from a denser medium to a rarer medium at an angle of incidence *i*. The reflected and refracted rays make an angle of 90° with each other. The angle of reflection and refraction are respectively *r* and *r′*. The critical angle is given by: **[22 July, 2021 (Shift-II)]**

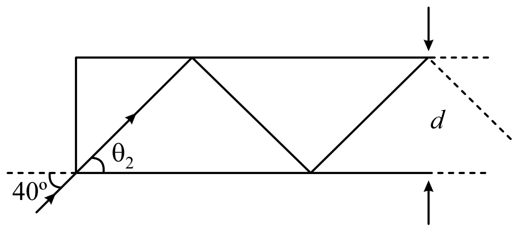


(*a*) $\sin^{-1}$ (cot *r*) (*b*) $\tan^{-1}$ (sin *i*)
(*c*) $\sin^{-1}$ (tan *r'*) (*d*) $\sin^{-1}$ (tan *r*)

13. In figure, the optical fiber is $l = 2$ m long and has a diameter of $d = 20$ μm. If a ray of light is incident on one end of the fiber at angle $\theta_1 = 40°$, the number of reflection it makes before emerging from the other end is close to:

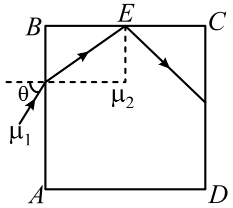
[8 April, 2019 (Shift-I)]

(refractive index of fibre is 1.31 and sin 40° = 0.64)



(*a*) 55000 (*b*) 57000
(*c*) 66000 (*d*) 45000

14. A transparent cube of side d, made of a material of refractive index μ_2, is immersed in a liquid of refractive index $\mu_1(\mu_1 < \mu_2)$. A ray is incident on the face AB at an angle θ (shown in the figure). Total internal reflection takes place at point E on the face BC. **[12 April, 2019 (Shift-II)]**

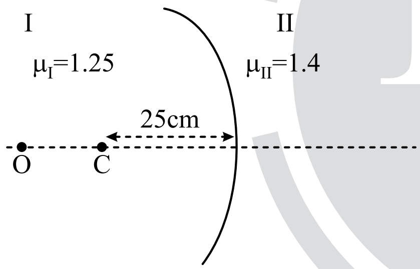


The θ must satisfy:

(*a*) $\theta < \sin^{-1}\dfrac{\mu_1}{\mu_2}$ (*b*) $\theta < \sin^{-1}\sqrt{\dfrac{\mu_2^2}{\mu_1^2}-1}$

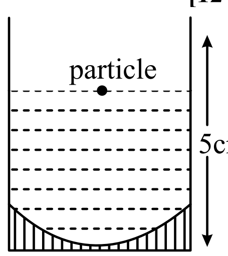
(*c*) $\theta > \sin^{-1}\dfrac{\mu_1}{\mu_2}$ (*d*) $\theta > \sin^{-1}\sqrt{\dfrac{\mu_2^2}{\mu_1^2}-1}$

15. Region I and II are separated by a spherical surface of radius 25 cm. An object is kept in region I at a distance of 40 cm from the surface. The distance of the image from the surface is: **[20 July, 2021 (Shift-I)]**



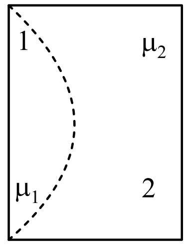
(*a*) 9.52 cm (*b*) 18.23 cm
(*c*) 37.58 cm (*d*) 55.44 cm

16. A concave mirror has radius of curvature of 40 cm. It is at the bottom of a glass that has water filled up to 5 cm (see figure). If a small particle is floating on the surface of water, its image as seen, from directly above the glass, is at a distance d from the surface of water. The value of d is close to: (Refractive index of water = 1.33) **[12 April, 2019 (Shift-I)]**



(*a*) 8.8 cm (*b*) 11.7 cm
(*c*) 6.7 mm (*d*) 13.4 cm

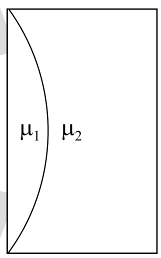
17. One plano-convex and one plano-concave lens of same radius of curvature 'R' but of different materials are joined side by side as shown in the figure. If the refractive index of the material of 1 is μ_1 and that of 2 is μ_2, then the focal length of the combination is: **[10 April, 2019 (Shift-I)]**



(*a*) $\dfrac{R}{2-(\mu_1-\mu_2)}$ (*b*) $\dfrac{2R}{\mu_1-\mu_2}$

(*c*) $\dfrac{R}{2(\mu_1-\mu_2)}$ (*d*) $\dfrac{R}{\mu_1-\mu_2}$

18. Curved surfaces of a plano-convex lens of refractive index μ_1 and a planoconcave lens of refractive index μ_2 have equal radius of curvature as shown in figure. Find the ratio of radius of curvature to the focal length of the combined lenses. **[27 Aug, 2021 (Shift-II)]**



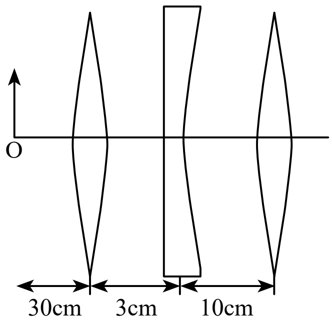
(*a*) $\dfrac{1}{\mu_2-\mu_1}$ (*b*) $\mu_2-\mu_1$

(*c*) $\dfrac{1}{\mu_1-\mu_2}$ (*d*) $\mu_1-\mu_2$

19. Find the distance of the image from object O, formed by the combination of lenses in the figure:

[27 Aug, 2021 (Shift-I)]

(*a*) 75 cm (*b*) 20 cm
(*c*) 10 cm (*d*) infinity



20. A microscope is focused on an object at the bottom of a bucket. If liquid with refractive index $\frac{5}{3}$ is poured inside the bucket, then microscope have to be raised by 30 cm to focus the object again. The height of the liquid in the bucket is: **[31 Jan, 2023 (Shift-II)]**

(*a*) 75 cm

(*b*) 50 cm

(*c*) 18 cm

(*d*) 12 cm

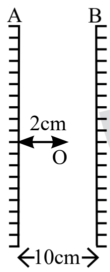
21. In normal adjustment, for a refracting telescope, the distance between objective and eye piece is 30 cm. The focal length of the objective, when the angular magnification of the telescope is 2 , will be:
[28 July, 2022 (Shift-I)]

(*a*) 20 cm (*b*) 30 cm

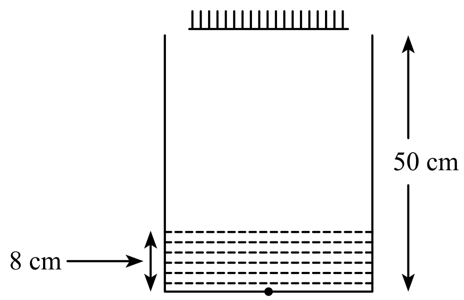
(*c*) 10 cm (*d*) 15 cm

Integer Type Questions

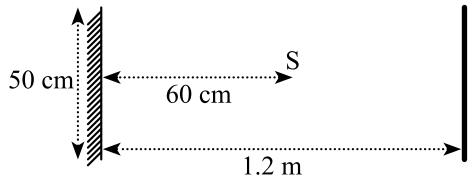
22. Two vertical parallel mirrors *A* and *B* are separated by 10 cm. A point object *O* is placed at a distance of 2 cm from mirror A. The distance of the second nearest image behind mirror *A* from the mirror *A* is ____ cm
[8 Apr, 2023 (Shift-I)]



23. As shown in the figure, a plane mirror is fixed at a height of 50 cm from the bottom of tank containing water $\left(\mu = \frac{4}{3}\right)$. The height of water in the tank is 8 cm. A small bulb is placed at the bottom of the water tank. The distance of image of the bulb formed by mirror from the bottom of the tank is _________ cm. **[11 Apr, 2023 (Shift-II)]**

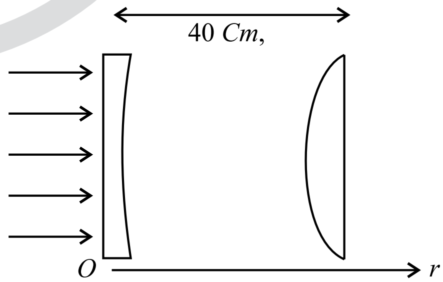


24. A point source of light S, placed at a distance 60 cm infront of the centre of a plane mirror of width 50 cm, hangs vertically on a wall. A man walks infront of the mirror along a line parallel to the mirror at a distance 1.2 m from it (see in the figure). The distance between the extreme points where he can see the image of the light source in the mirror is _____ cm. **[26 Feb, 2021 (Shift-II)]**



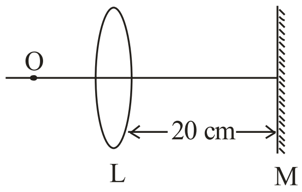
25. A small bulb is placed at the bottom of a tank containing water to a depth of $\sqrt{7}$ m. The refractive index of water is $\frac{4}{3}$. The area of the surface of water through which light from the bulb can emerge out is $x\pi$ m^2. The value of x is ________. [26 June, 2022 (Shift-II)]

26. As shown in the figure, a combination of a thin plano concave lens and a thin plano convex lens is used to image an object placed at infinity. The radius of curvature of both the lenses is 30 *cm* and refractive index of the material for both the lenses is 1.75. Both the lenses are placed at distance of 40 *cm* from each other. Due to the combination, the image of the object is formed at distance x = _________ *cm*, from concave lens. **[24 Jan, 2023 (Shift-I)]**

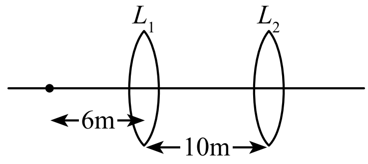


27. Two transparent media having refractive indices 1.0 and 1.5 are separated by a spherical refracting surface of radius of curvature 30 cm. The centre of curvature of surface is towards denser medium and a point object is placed on the principle axis in rarer medium at a distance of 15 cm from the pole of the surface. The distance of image from the pole of the surface is _______ cm.
[8 Apr, 2023 (Shift-II)]

28. An object is placed on the principal axis of convex lens of focal length 10 cm as shown. A plane mirror is placed on the other side of the lens at a distance of 20 cm. The image produced by the plane mirror is 5 cm inside the mirror. The distance of the object from the lens is ______ cm. **[25 Jan, 2023 (Shift-II)]**



29. A point object, 'O' is placed in front of two thin symmetrical coaxial convex lenses L_1 and L_2 with focal length 24 cm and 9 cm respectively. The distance between two lenses is 10 cm and the object is placed 6 cm away from lens L_1 as shown in the figure. The distance between the object and the image formed by the system of two lenses is _____cm. **[10 Apr, 2023 (Shift-II)]**

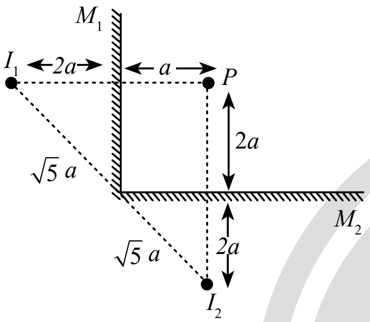


ANSWER KEY

1. (*c*)	**2.** (*c*)	**3.** (*c*)	**4.** (*d*)	**5.** (*b*)	**6.** (*a*)	**7.** (*d*)	**8.** (*d*)	**9.** (*c*)	**10.** (*d*)
11. (*b*)	**12.** (*d*)	**13.** (a, b)	**14.** (*b*)	**15.** (*c*)	**16.** (*a*)	**17.** (*d*)	**18.** (*d*)	**19.** (*a*)	**20.** (*a*)
21. (*a*)	**22.** [18]	**23.** [98]	**24.** [150]	**25.** [9]	**26.** [120]	**27.** [30]	**28.** [30]	**29.** [34]	

EXPLANATIONS

1. (*c*)



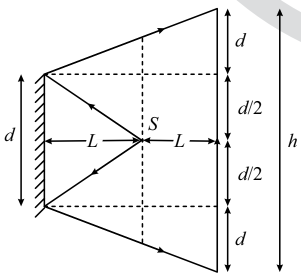
The shortest distance between the image thus formed,
$I_1I_2 = 2\sqrt{5}a$

$= 2 \times 2.3\ a = 4.6\ a$

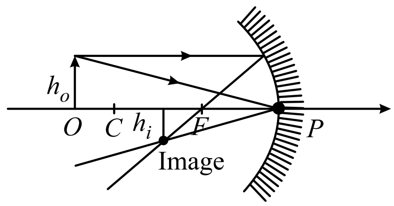
2. (*c*) So, the final distance is

$$h = d + \frac{d}{2} + \frac{d}{2} + d = 3d.$$

$$h = d + d + d = 3d.$$



3. (*c*)



$h_i < h_o$

Therefore, the image formed is real, inverted and diminished or unmagnified.

Hence, option (*c*) is correct answer

4. (*d*) At focus, magnification is ∞.

5. (*b*) From the formula $n = \dfrac{\text{real depth}}{\text{apparent depth}}$

$$d_{\text{app}} = \frac{d_1}{n_1} + \frac{d_2}{n_2}$$

$$d_{\text{app}} = \frac{d}{2}\left[\frac{n_1 + n_2}{n_1 n_2}\right]$$

6. (*a*) Since, $\mu = \dfrac{c}{V}$, therefore $\mu \propto \dfrac{1}{v}$

$$\frac{m_A}{m_B} = \frac{v_B}{v_A} = \frac{1}{2}$$

Let the thickness is *d*

Given: $t_1 - t_2 = 5 \times 10^{-10}$

$$\frac{d}{v_B} - \frac{d}{v_A} = 5 \times 10^{-10}$$

$$d = \frac{5 \times 10^{-10} \times v_A v_B}{v_A - v_B}$$

As $v_A = 2v_B \Rightarrow d = \dfrac{5 \times 10^{-10} \times 2v_A . v_B}{2v_A - v_B} = 5 \times 10^{-10} \times 2V_B$

or $d = 5 \times 10^{-10} \times v_A$

7. (*d*) From Snell's law

$$\sin i \times 1 = \sin\frac{i}{2} \times \sqrt{2n}$$

$$2\sin\frac{i}{2}\cdot\cos\frac{i}{2}=\sin\frac{i}{2}\times\sqrt{2n}$$

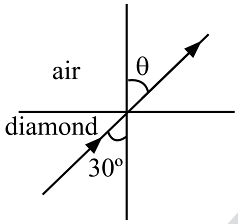
$$\cos\frac{i}{2}=\sqrt{\frac{n}{2}}\Rightarrow i=2\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$$

8. (*d*) From snell's law

$\mu_1 \sin 30^\circ = \mu_2 \sin\theta$

$\Rightarrow 2.42\times\frac{1}{2}=(1)\sin\theta$

$\sin\theta = 1.21 > 1$



Refraction is not possible

9. (*c*) Apparent depth of the inner surface of the bottom of the vessel, $d=\frac{h}{\sqrt{2}}+\frac{h}{2\sqrt{2}}$

$\Rightarrow d=\frac{h}{\sqrt{2}}\times\frac{3}{2}=\frac{3\sqrt{2}\,h}{4}$

10. (*d*) We have given,

$v_A = 2\times 10^{10}$ cm/s, $v_B = 1.5\times 10^{10}$ cm/s

We know that, for total internal reflection,

$$\frac{\sin\theta_C}{v_B}=\frac{\sin 90^\circ}{v_A}\Rightarrow \sin\theta_C=\frac{v_B}{v_A}=\frac{1.5\times10^{10}}{2.0\times10^{10}}=\frac{3}{4}$$

$$\Rightarrow \theta_C=\sin^{-1}\left(\frac{3}{4}\right)$$

According to question, we can write

$$\theta>\theta_c=\sin^{-1}\left(\frac{3}{4}\right)$$

11. (*b*) Applying snell's law on first surface

$$1\times\sin\theta=\frac{4}{3}\times\sin\theta'\Rightarrow\sin\theta'=\frac{3}{4}\sin\theta \;...\text{(i)}$$

For TIR on second surface

$\sin\theta'' > \sin\theta_c$ $(\theta' + \theta'' = 90)$

$\sin(90 - \theta') > \sin\theta_c$

$\cos\theta' > \sin\theta_c$

$\cos^2\theta' > \sin^2\theta_c \Rightarrow 1 - \sin^2\theta' > \sin^2\theta_c$

$$1-\left(\frac{3}{4}\sin\theta\right)^2>\sin^2\theta_c \quad \text{(By equation (i))}$$

$$1-\frac{9}{16}\sin^2\theta>\sin^2\theta_c \;\therefore\left[\sin\theta_c=\frac{1}{\mu}=\frac{3}{4}\right]$$

$$1-\frac{9}{16}\sin^2\theta>\frac{9}{16}\Rightarrow 1-\frac{9}{16}>\frac{9}{16}\sin^2\theta$$

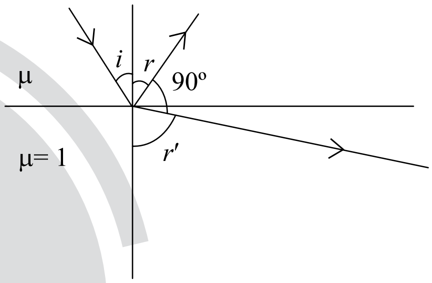
$$\frac{7}{16}>\frac{9}{16}\sin^2\theta\Rightarrow\sin\theta<\frac{\sqrt{7}}{3}$$

12. (*d*) $r + r' = 90^\circ$

$\Rightarrow r' = 90^\circ - r$

$\because$ Law of refraction $i = r$

$\therefore r' = 90^\circ - i$



Snell's law:

$\mu \sin i = 1 \sin r' \Rightarrow \mu \sin i = \sin(90^\circ - i)$

$\Rightarrow \tan i=\frac{1}{\mu}=\tan r$

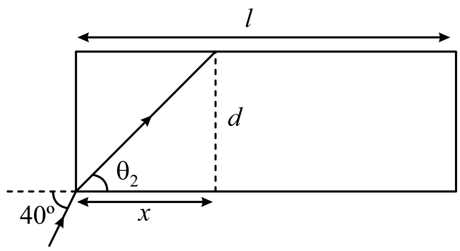
Now critical angle:

$\mu \sin C = 1 \sin 90^\circ = 1$

$\Rightarrow \sin C=\frac{1}{\mu}=\tan r$

$C = \sin^{-1}(\tan r)$

13.(a, *b*)



Exact solution

by Snells' law 1.sin 40° = (1.31) sin θ_2

$$\sin\theta_2=\frac{0.64}{1.31}=\frac{64}{131}\approx 0.49$$

$$\text{Now }\tan\theta_2=\frac{64}{\sqrt{(131)^2-(64)^2}}$$

$$\frac{64}{\sqrt{13065}} \approx \frac{64}{114.3} = \frac{d}{x}$$

Now no. of reflections $= \frac{2}{x}$

$$= \frac{2\times 64}{114.3\times 20\times 10^{-6}} = \frac{64\times 10^{5}}{114.3}$$

$\approx 55991 \approx 55000$

Approximate solution

By Snells' law 1. sin 40° = (1.31) sin θ_2

$$\sin\theta_2 = \frac{0.64}{1.31} = \frac{64}{131} \approx 0.49$$

If assume $\Rightarrow \theta_2 \approx 30°$

$$\tan 30° = \frac{d}{x} \Rightarrow x = \sqrt{3}\,d$$

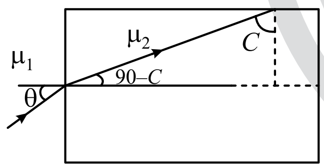
Now number of reflections

$$= \frac{1}{\sqrt{3}d} = \frac{2}{\sqrt{3}\times 20\times 10^{-6}} = \frac{10^5}{\sqrt{3}}$$

$\approx 57735 \approx 57000$

Note : If we approximate the angle θ_2 as 30° initially then answer will be closer to 57000. But if we solve thoroughly, answer will be close to 55000. So both the answers must be awarded. Detailed solution is as following.

14. (*b*)



$$\sin c = \frac{\mu_1}{\mu_2}$$

$$\mu_1 \sin\theta = \mu_2 \sin(90° - C) = \mu_2 \cos C$$

$$\sin\theta = \frac{\mu_2\sqrt{1-\frac{\mu_1^2}{\mu_2^2}}}{\mu_1}$$

$$\theta = \sin^{-1}\sqrt{\frac{\mu_2^2-\mu_1^2}{\mu_1^2}}$$

For TIR

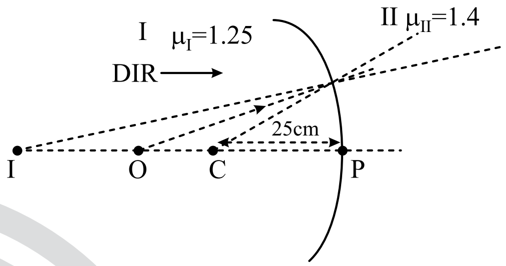
$$\theta < \sin^{-1}\sqrt{\frac{\mu_2^2}{\mu_1^2}-1}$$

15. (*c*) $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2-\mu_1}{R}$

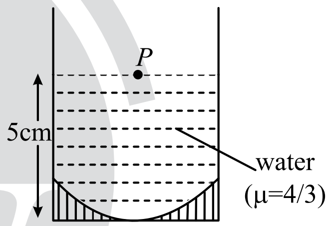
$$\Rightarrow \frac{1.4}{v} - \frac{1.25}{-40} = \frac{1.4-1.25}{-25}$$

$$\Rightarrow -\frac{1.4}{v} = \frac{0.15}{25} + \frac{1.25}{40} = \frac{7.45}{200}$$

$$\Rightarrow v = -\frac{200\times 1.4}{7.45} = -37.58\text{cm}$$



16. (*a*) Light incident from particle P will be reflected at mirror



$$u = -5\text{cm},\ f = -\frac{R}{2} = -20\text{ cm}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$v_1 = +\frac{20}{3}\text{cm}$$

This image will act as object for light getting refracted at water surface

So, object distance $d = 5 + \frac{20}{3} = \frac{35}{3}$ cm

below water surface.

After refraction, final image is at

$$d' = d\left(\frac{\mu_2}{\mu_1}\right)$$

$$= \left(\frac{35}{3}\right)\left(\frac{1}{4/3}\right)$$

$$= \frac{35}{4} = 8.75\text{ cm} \approx 8.8\text{ cm}$$

17. (*d*) For 1st lens $\dfrac{1}{f_1}=\left(\dfrac{\mu_1-1}{1}\right)\left(\dfrac{1}{\infty}-\dfrac{1}{-R}\right)=\dfrac{\mu_1-1}{R}$

for 2nd lens $\dfrac{1}{f_2}=\left(\dfrac{\mu_2-1}{1}\right)\left(\dfrac{1}{-R}-0\right)=-\dfrac{\mu_2-1}{R}$

$$\frac{1}{f_{eq}}=\frac{1}{f_1}+\frac{1}{f_2}$$

$$\frac{1}{f_{eq}}=\frac{\mu_1-1}{R}-\frac{\mu_2-1}{R}=\frac{\mu_1-\mu_2}{R}$$

$$\Rightarrow f_{eq}=\frac{R}{\mu_1-\mu_2}$$

18. (*d*) $\dfrac{1}{f_1}=\dfrac{\mu_1-1}{R}$

$$\frac{1}{f_2}=-\left(\frac{\mu_2-1}{R}\right)$$

$$\frac{1}{f_{eq}}=\frac{1}{f_1}+\frac{1}{f_2}=\frac{\mu_1-\mu_2}{R}\quad\Rightarrow\frac{R}{f_{eq}}=\mu_1-\mu_2$$

19. (*a*) $\dfrac{1}{v_1}+\dfrac{1}{30}=\dfrac{1}{10}$

$$\Rightarrow v_1=15\text{cm}$$

$$\Rightarrow\frac{1}{v_2}-\frac{1}{10}=-\frac{1}{10}$$

$$\Rightarrow v_2=\infty\Rightarrow v_3=30\text{cm}\Rightarrow OV_3=75\text{cm}$$

20. (*a*) Using formula, $d=h\left(1-\dfrac{1}{\mu}\right)$

$$\Rightarrow\ 30=h\left(1-\frac{3}{5}\right)\qquad\left[\because\text{Given, }d=30\text{cm},\mu=\frac{5}{3}\right]$$

$$\Rightarrow\ 30=h\times\frac{2}{5}$$

$$\therefore\ h=75\text{ cm}$$

21. (*a*) $|f_0|+|f_e|=30$ cm (Given)

As, $\left|\dfrac{f_0}{f_e}\right|=2$ (given)

$$\Rightarrow|f_0|=2\,|f_e|$$

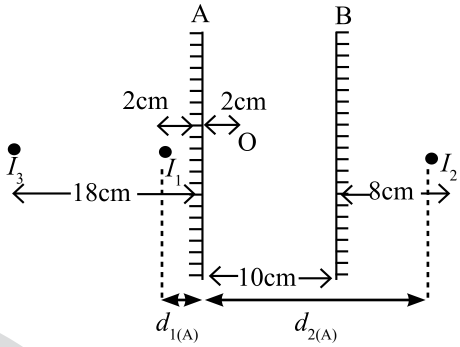
$$\therefore|f_0|+\frac{|f_0|}{2}=30\Rightarrow|f_0|=\frac{2}{3}\times30=20\,cm$$

22. [18] I_1 is the first nearest image to A.

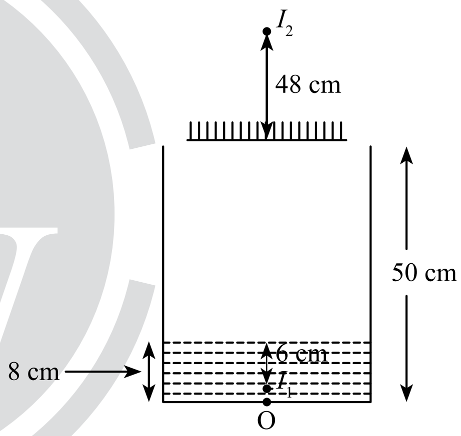
$d_{1(A)}=2$ cm

I_2 is the second nearest image to A.

$d_{2(A)}=18$ cm (Desired)



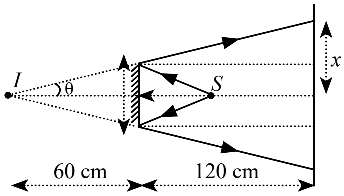
23. [98]



Apparent depth of $O=\dfrac{\text{real depth}}{\text{refractire index}}=\dfrac{d}{\mu}=6$

Distance between O and $I_2=48+50=98$ cm

24. [150]

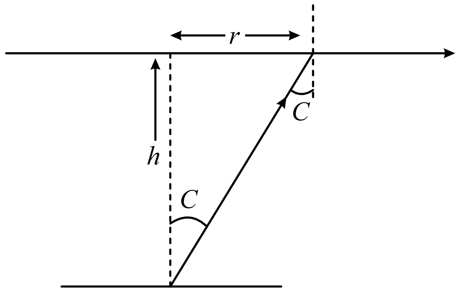


$$\tan\theta=\frac{25}{60}=\frac{x}{180}$$

$x=75$ cm

$\Rightarrow$ Maximum Distance $=2x=2\times75=150$ cm

25. [9] Given, $h = \sqrt{7}m$, $\mu = 4/3$



$\tan C = \dfrac{r}{h}$

$r = h \tan C$

$\sin C = \dfrac{1}{\mu} = \dfrac{3}{4}$

$\tan C = \dfrac{3}{\sqrt{7}}$

$r = \sqrt{7} \times \dfrac{3}{\sqrt{7}} = 3$

Area of surface = $\pi r^2 = 9\pi$ m^2

26. [120] For the plano concave lens, $\dfrac{1}{f_1} = (1.75-1)\left(-\dfrac{1}{30}\right)$

$\Rightarrow f_1 = -40\,cm$

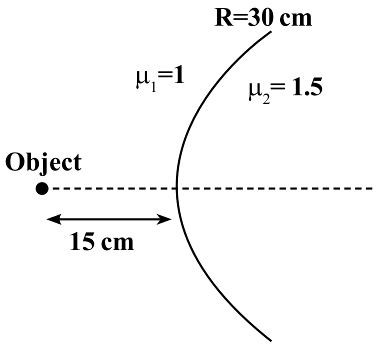
For the plano convex lens,

$\dfrac{1}{f_2} = (1.75-1)\left(\dfrac{1}{30}\right) \Rightarrow f_2 = 40\,cm$

Image from lens L_1 will be virtual and to the left of L_1 at focal length 40 *cm*. So the object for lens L_2 will be 80 *cm* from L_2 which is equal to 2*f*. Final image is formed at 80 *cm* from lens L_2 on the right side of the lens.

$\Rightarrow x = 120$

27. [30]



$\dfrac{\mu_2}{v} - \dfrac{\mu_1}{u} = \dfrac{\mu_2 - \mu_1}{R}$

$\dfrac{1.5}{v} - \dfrac{1}{(-15)} = \dfrac{1.5-1}{30}$

$\dfrac{1.5}{v} + \dfrac{1}{15} = \dfrac{1}{60} \quad \Rightarrow v = -30$ cm

28. [30]

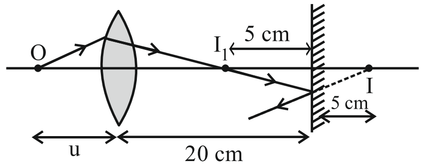


Image distance,

$v = 20 - 5$

$= 15$ cm

$\dfrac{1}{v} - \dfrac{1}{u} = \dfrac{1}{f}$

$\dfrac{1}{15} - \dfrac{1}{-u} = \dfrac{1}{10}$

$\Rightarrow \dfrac{1}{u} = \dfrac{1}{10} - \dfrac{1}{15}$

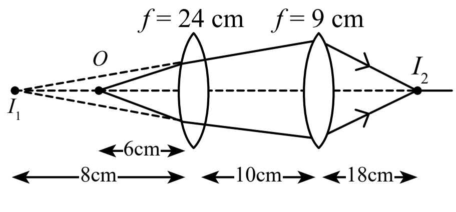
$\Rightarrow u = 30$cm.

29. [34] For $L_1 : \dfrac{1}{v} - \dfrac{1}{-6} = \dfrac{1}{24}$

$\dfrac{1}{v} = \dfrac{1}{24} - \dfrac{1}{6} \Rightarrow v = -8$cm

For $L_2 : \dfrac{1}{v} - \dfrac{1}{(-8-10)} = \dfrac{1}{9}$

$\dfrac{1}{v} = \dfrac{1}{9} - \dfrac{1}{18} \Rightarrow v = 18$



So $OI_2 = 6 + 10 + 18 = 34$ cm